Generation Depth Fingerprinting: Classifying Recursive LLM Output Beyond Binary Detection

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Abstract

Web-scraped training corpora increasingly contain LLM-generated text of unknown provenance depth—material that has been recursively generated one or more times. Current detectors offer only binary real-vs-synthetic classification, conflating qualitatively different generations. We formulate **generation depth classification** (gen-0 / gen-1 / gen-2+) and characterize **differential depth response** (DDR): central-tendency features (entropy mean, TTR, hapax ratio) drift monotonically with depth ($|\rho| > 0.60$), while distributional shape features (entropy max, tail decay) remain near-stationary ($|\rho| < 0.10$), with a $\sim 10\times$ gap in KS statistics between groups. A lightweight 7-feature ensemble achieves leak-free binary F1 = 0.694 (95% CI [0.688, 0.700]) under 8-generator leave-one-generator-out (LOGO) evaluation across 5 model families (0.6B–8B), and 0.830 on cross-model chains. DDR is robust across tested decoding strategies (greedy, top-k) and three reference models from different families. We show that depth-aware data curation is statistically equivalent to random subsampling for downstream CLM quality (TOST $p = 1.1 \times 10^{-4}$ at $\pm 2\%$ margin), establishing depth fingerprinting as a *diagnostic* tool for provenance characterization rather than a curative one for data selection. The signal degrades substantially under instruction tuning, restricting current scope to base models.

1 Introduction

Large language models (LLMs) are increasingly deployed at scale, and their outputs feed back into the training pipelines of subsequent models. Recent estimates suggest 7–17% of scientific papers may contain LLM-generated content, and web crawls used for pretraining are progressively contaminated with synthetic text of varying provenance. This recursive contamination—where model-generated text is regenerated by the same or different models—raises fundamental questions about data quality and the distributional properties of recursively generated text.

Current AI-text detection methods frame the problem as binary classification: human-written vs. machine-generated. This conflation obscures meaningful differences. First-generation text (gen-1) retains much of the source distribution's lexical properties, while deeper generations (gen-2+) exhibit measurably reduced vocabulary diversity and compressed perplexity distributions. Existing detectors such as FastDetectGPT (**?**), Binoculars (**?**), and LD-Score cannot distinguish these qualitatively different materials.

We move beyond binary detection and formulate **generation depth classification**: given a text passage and a reference language model, estimate whether the text is original (gen-0), first-generation synthetic (gen-1), or deeper-generation synthetic (gen-2+). Our central empirical finding is **differential depth response** (DDR): features measuring central tendency and lexical diversity drift monotonically with generation depth, while features capturing distributional shape remain approximately stationary. This is consistent with recursive generation inducing a location shift in the NLL distribution while preserving its shape.

Our contributions are:

1. **Task formulation**: Generation depth classification as a new problem beyond binary detection, with a rigorous LOGO evaluation protocol that prevents generator leakage.
2. **Empirical characterization of DDR** across 5 generators from 3 families (0.6B–8B), validated with 3 reference models. DDR is robust across tested decoding strategies (greedy, top-k).
3. **Negative finding**: Depth-aware curation does not improve downstream CLM quality over random subsampling (TOST-confirmed equivalence), positioning depth fingerprinting as diagnostic rather than curative.

2 Related Work

AI-Text Detection. Perplexity-based detectors leverage the observation that model-generated text has lower perplexity under the generating model. GLTR (**?**) visualizes token-level statistics; DetectGPT (**?**) uses perturbation-based log-probability curvature; FastDetectGPT (**?**) replaces perturbations with conditional sampling for efficiency; Binoculars (**?**) contrasts two models' cross-entropy scores. All produce binary decisions and do not estimate generation depth. When repurposed as ordinal depth predictors (using continuous scores as features), these methods show high cross-generator variance and substantially lower performance than multi-feature approaches.

Model Collapse and Iterative Generation. ? show that training on model-generated data leads to "model collapse" with progressive distribution narrowing. The "LLM as Broken Telephone" framework (ACL 2025) demonstrates that paraphrasing chains converge to attractor states within $\sim$3 iterations under semantic-preservation constraints. Our work differs in studying non-paraphrastic continuation chains, where the absence of semantic constraints permits monotonic drift beyond the attractor horizon.

Data Quality and OOD Detection. Perplexity-based data scoring is standard in pretraining pipelines. Our reference-model features (entropy statistics, tail decay) are related to OOD detection methods that use NLL distribution properties. The distinction is that we exploit the *ordinal* structure of depth-induced distributional shift rather than binary in-vs-out classification.

Authorship Attribution and Stylometry. Surface features such as TTR and hapax legomena ratio have long histories in authorship attribution. We repurpose these as depth indicators, showing they capture vocabulary impoverishment across recursive generation.

3 Method

3.1 Problem Formulation

Given a text passage x and a reference model M_{ref}, we classify x into one of three depth classes: gen-0 (original human text), gen-1 (first-generation LLM output), or gen-2+ (deeper recursive generations). We define **generation depth** operationally: depth d text is produced by feeding depth $d-1$ text as a prompt to a generator model, starting from human-written seeds at $d=0$.

3.2 Feature Design

We extract 7 features per passage, divided into two groups motivated by the DDR hypothesis:

Surface features (2): Type-Token Ratio (TTR) and hapax legomena ratio. These capture vocabulary diversity, which decreases monotonically with generation depth as models converge toward high-frequency token distributions.

Reference features (5): Given reference model M_{ref}, we compute token-level negative log-likelihood (NLL) values and extract: entropy mean, entropy standard deviation, entropy minimum, entropy maximum, and tail decay rate. These capture properties of the NLL distribution that respond differentially to generation depth.

3.3 Differential Depth Response (DDR)

We classify features into three groups based on their Spearman correlation ρ with depth and KS test statistics:

- **Drifting** ($|\rho| > 0.50$, KS > 0.50): hapax ratio ($\rho = -0.63$), TTR (-0.61), entropy mean (-0.60).
- **Moderate** ($0.10 < |\rho| < 0.50$): entropy min (-0.35), entropy std (-0.12).
- **Near-stationary** ($|\rho| < 0.10$, KS < 0.06): entropy max (0.06), tail decay (0.04).

The $\sim 10\times$ gap in KS statistics between drifting features (0.516–0.537) and near-stationary features (0.043–0.052) provides non-parametric confirmation of the three-tier DDR structure.

3.4 Classifier

We train an XGBoost classifier on the 7-feature vector. Evaluation follows the leave-one-generator-out (LOGO) protocol: models are trained on chains from $N-1$ generators and tested on the held-out generator, ensuring no generator-specific artifacts leak into evaluation. The primary metric is **leak-free binary F1** (gen-1 vs. gen-2+), excluding gen-0 to avoid seed-sharing leakage between gen-0 samples across generators.

3.5 Chain Construction

We construct recursive generation chains of depth 0–4 using three prompt types:

- **Continuation**: Direct text continuation from the previous depth's output.

Feature Set	Macro F1	95% CI
Full (7 features)	0.820	[0.807, 0.833]
Surface-only (TTR + hapax)	0.763	[0.747, 0.774]
Reference-only (5 entropy + tail)	0.719	[0.706, 0.733]

Table 1: Feature ablation (5-fold CV, Pythia-2.8B). McNemar's test confirms Full $>$ Surface $>$ Reference (all $p \approx 0$).

Feature	Spearman ρ	KS	DDR Group
hapax_ratio	-0.63	0.537	drifting
TTR	-0.61	0.516	drifting
entropy_mean	-0.60	0.528	drifting
entropy_min	-0.35	0.052	moderate
entropy_std	-0.12	0.043	moderate
entropy_max	$+0.06$	0.048	near-stationary
tail_decay	$+0.04$	0.045	near-stationary

Table 2: DDR feature classification. Drifting features show strong monotonic depth correlation and high distributional shift; near-stationary features show negligible response.

- **Elaboration**: Prompted to elaborate on the previous depth's content.
- **Inspired rewriting**: Prompted to rewrite inspired by the previous content.

Seeds are drawn from multiple domains (Wikipedia, news, academic, code) with 500–1000 seeds per domain.

4 Experiments

4.1 Setup

We evaluate on 5 base-model generators spanning 3 families: Pythia-2.8B, Pythia-1B (Pythia family), OLMo-2-1B (OLMo), Qwen3-1.7B, and Qwen3-0.6B (Qwen). An additional large-scale experiment includes Qwen3-8B. The default reference model is Pythia-410M; cross-reference experiments use OLMo-2-1B and Qwen3-0.6B as alternative references. Total dataset: 18,423 passages across 5 generators.

4.2 Feature Ablation and DDR Analysis

Table 1 shows the feature ablation results on Pythia-2.8B using 5-fold cross-validation.

The full feature set significantly outperforms both subsets, confirming feature complementarity. Surface features contribute more individually (+4.4pp over reference-only), but reference features provide independent complementary signal (+5.7pp when added to surface).

Table 2 presents the DDR feature classification. Drifting features exhibit KS statistics 0.516–0.537, while near-stationary features show 0.043–0.052—a $\sim 10\times$ gap that provides non-parametric confirmation of DDR.

Median perplexity decreases monotonically with depth (13.50, 4.96, 3.00, 2.07, 1.72 for depths 0–4), validating the location-shift interpretation: recursive generation progressively moves the NLL distribution toward lower values.

4.3 Cross-Generator Evaluation

Table 3 reports per-generator results under 5-generator LOGO evaluation.

Generator	Family	Size	Bin. F1	3-cl. F1
Pythia-2.8B	Pythia	2.8B	0.724	0.812
Pythia-1B	Pythia	1B	0.721	0.851
Qwen3-1.7B	Qwen	1.7B	0.688	0.811
OLMo-2-1B	OLMo	1.5B	0.662	0.752
Qwen3-0.6B	Qwen	0.6B	0.656	0.819
Average			**0.694**	**0.804**

Table 3: LOGO per-generator results. Binary F1 = gen-1 vs gen-2+ (leak-free). Per-generator range: 0.656–0.724 (6.8pp spread).

The LOGO binary F1 of 0.694 [0.688, 0.700] demonstrates that the depth signal generalizes to unseen generators. The 6.8pp per-generator spread (0.656–0.724) reflects generator-specific variation in depth signal strength. Feature importance is stable across generators: hapax ratio (0.289), TTR (0.168), entropy mean (0.160).

Cross-family transfer. Leave-one-family-out (LOFO) evaluation confirms robust cross-family generalization (Table 4). The average LOFO binary F1 of 0.686 is only 0.8pp below LOGO, indicating that depth features transfer across model families.

4.4 Cross-Reference Model Validation

To verify DDR is not an artifact of the Pythia-410M reference, we replicate with OLMo-2-1B and Qwen3-0.6B as reference models (Table 5).

All three references yield consistent DDR: entropy drift is uniformly negative, tail decay $|\rho|$ remains below 0.07, and LOGO 3-class F1 ranges 0.803–0.816 (1.3pp spread). Greedy decoding shape stability is confirmed across all references (mean $|d|$ = 0.05–0.06). While drift magnitude varies by reference (OLMo reference drift is $\sim 50\%$ of Pythia's), the qualitative DDR structure—drifting vs. near-stationary—is preserved.

Held-out Family	Binary F1	Δ vs LOGO
Pythia	0.678	−0.016
OLMo	0.691	−0.003
Qwen	0.689	−0.005
Average	**0.686**	−0.008

Table 4: LOFO cross-family transfer. Delta vs LOGO is minimal (−0.8pp average).

Reference	Size	3-cl. F1	ent. drift	|tail ρ|
Pythia-410M	410M	0.803	[−1.19, −1.40]	0.05
OLMo-2-1B	1.5B	0.807	[−0.61, −0.91]	0.07
Qwen3-0.6B	0.6B	0.816	[−0.96, −1.35]	0.04

Table 5: Reference model comparison. DDR direction is consistent across all references; LOGO F1 varies only 1.3pp. Drift magnitude is reference-dependent (OLMo ∼50% of Pythia), but the drifting/stationary classification is preserved.

4.5 Baseline Comparison

Table 6 compares the 7-feature ensemble against existing detectors repurposed as ordinal depth predictors.

The 7-feature ensemble outperforms the best single detector (Avg-LogProb) by 3.3pp ($p < 0.001$, bootstrap). More importantly, its CI [0.786, 0.799] is far narrower than Avg-LogProb's [0.616, 0.820], demonstrating consistent performance across generators. Single detectors designed for binary classification show high cross-generator variance when repurposed for ordinal depth prediction—their ordinal utility is limited.

4.6 Downstream Utility: A Negative Finding

We test whether depth-aware data curation improves downstream language modeling quality by fine-tuning OLMo-1B on depth-stratified subsets (3 seeds each). Table 7 reports perplexity results.

Depth-aware curation does not improve downstream CLM quality. At equal sample count ($n = 3{,}951$), random subsampling (PPL = 2.367) is statistically equivalent to depth 0+1 filtering (PPL = 2.387; TOST $p = 1.1 \times 10^{-4}$ at $\pm 2\%$ margin). The depth-0-only variant (PPL = 2.681) is substantially worse, but this reflects sample size reduction (only 797 texts) rather than depth quality effects. Seed variance is minimal (< 0.002), confirming reproducibility.

This negative finding is the most robust of our claims: depth fingerprinting is a *diagnostic* tool for provenance characterization, not a curative tool for data curation.

Method	F1	95% CI	ρ	Type
7-feat ensemble	0.798	[.786, .799]	—	multi
Avg-LogProb	0.765	[.616, .820]	−0.79	single
LD-Score	0.721	[.588, .770]	−0.79	single
FastDetectGPT	0.627	[.511, .762]	−0.43	single
Binoculars	0.579	[.405, .664]	−0.20	single

Table 6: Ordinal baseline comparison (LOGO 3-class F1). The 7-feature ensemble outperforms all single detectors (+3.3pp vs best, $p < 0.001$) with substantially narrower CI, reflecting robust cross-generator behavior.

Variant	n	PPL	TOST p
All depths	12,299	2.335 ± 0.000	—
Size-matched random	3,951	2.367 ± 0.000	ref
Depth 0+1 only	3,951	2.387 ± 0.001	1.1×10^{-4}
Binary-filtered	3,951	2.389 ± 0.001	$\sim10^{-4}$
Depth 0 only	797	2.681 ± 0.001	reject

Table 7: Downstream CLM quality by training data composition. TOST at ±2% margin confirms statistical equivalence between size-matched random subsampling and depth-filtered variants. Depth 0 only is worse due to sample size, not depth quality.

4.7 Prompt-Type Analysis

Table 8 reports performance by prompt type.

Continuation chains yield the strongest depth signal (CV F1 = 0.781), while elaboration is weakest (0.730)—a 5.1pp gap reflecting moderate prompt-type dependence. The leave-one-prompt-type-out (LOPTO) average F1 of 0.753 confirms reasonable cross-prompt transfer.

4.8 Paraphrastic Comparison

To connect with attractor cycle theory, we compare continuation and paraphrastic chains (Table 9).

Paraphrastic chains yield binary F1 of only 0.572 ($\Delta = -12.3$pp vs continuation), with hapax $|\rho|$ dropping from 0.63 to 0.24. The semantic-preservation constraint locks paraphrastic distributions onto an attractor manifold by ∼depth 2.5, consistent with prior work on iterative paraphrasing. This confirms that our DDR characterization applies specifically to non-paraphrastic chains where the absence of semantic constraints permits sustained monotonic drift.

4.9 Cross-Model Chains

To simulate realistic web contamination where different generators contribute at each depth step, we construct cross-model chains (Table 10).

Cross-model chains yield binary F1 = 0.830 under generator-level LOGO, +13.6pp over single-

Prompt	CV F1	Ret.	hapax ρ	ent. ρ
Continuation	0.781	0.831	-0.74	-0.56
Rewriting	0.737	0.939	-0.80	-0.48
Elaboration	0.730	0.986	-0.83	-0.47

Table 8: Prompt-type comparison (Pythia-2.8B). Ret. = depth retention (fraction of chains reaching depth 4). Continuation shows strongest classification signal; elaboration preserves more chains but with weaker signal.

Chain	Bin. F1	3-cl.	hpx ρ	ent ρ	Conv.
Contin.	0.695	0.762	-0.63	-0.60	$>$d4
Paraph.	0.572	0.580	-0.24	-0.12	$\sim$d2.5

Table 9: Paraphrastic vs continuation chains. Conv. = approximate attractor convergence depth. Paraphrastic chains converge to attractor states by $\sim$depth 2.5, severely attenuating depth signal.

model LOGO. Architecture differences at each depth step introduce additional distributional shifts that amplify the depth signal. The composition-level LOGO (0.764) is lower, indicating that specific generator combinations do not fully transfer.

4.10 Multi-Domain Analysis

Table 11 shows binary F1 across four seed domains.

Wikipedia yields the highest F1 (0.815), likely due to distributional alignment with pretraining data. Academic text is lowest (0.739) because its inherently specialized vocabulary compresses the feature margin. The cross-domain LOGO (0.700) is close to the overall LOGO (0.694), confirming moderate domain independence.

4.11 Finer-Grained Ordinal Classification

Table 12 tests whether finer granularity is achievable.

Five-class classification yields F1 $\sim$0.50, and ordinal methods (CORAL, Frank-Hall) provide no improvement over standard XGBoost. Gen-0 is easily identified (F1 = 0.95), but deeper classes are mutually confusable: gen-4 F1 is only 0.28. The bottleneck is feature resolution, not classifier architecture—DDR feature drift slopes flatten after depth 2, making adjacent deep generations indistinguishable. This establishes 3-class (gen-0/gen-1/gen-2+) as the practical resolution ceiling, with principled boundaries aligned to feature discriminability.

Evaluation	Bin. F1	95% CI	hpx ρ
Cross-model (gen-LOGO)	0.830	[.815, .845]	-0.68
Cross-model (comp-LOGO)	0.764	[.745, .783]	-0.65
Single-model (LOGO)	0.694	[.688, .700]	-0.63

Table 10: Cross-model vs single-model chains. Architecture heterogeneity amplifies depth signal (+13.6pp binary F1), partly attributable to architecture-mixing effects layered on top of depth-intrinsic drift.

Domain	Binary F1	95% CI	Seeds
Wikipedia	0.815	[0.798, 0.832]	1000
Code	0.754	[0.737, 0.770]	500
News	0.746	[0.726, 0.764]	500
Academic	0.739	[0.720, 0.757]	500
Cross-domain LOGO	0.700	—	—

Table 11: Multi-domain binary F1. Max gap vs Wikipedia is 7.6pp. Cross-domain LOGO (0.700) confirms depth signal is domain-robust.

4.12 Scale Effect

Adding Qwen3-8B (Table 13) tests whether depth signal persists in larger generators.

Qwen3-8B held-out binary F1 = 0.683—lower than smaller generators but still well above random (0.50). The DDR three-tier structure is qualitatively preserved: hapax (-0.56) and TTR (-0.54) remain strongly drifting, while tail decay (-0.09) stays near-stationary. Surface features are relatively more robust to scale (hapax $|\rho|$ drops 11%) than reference features (entropy mean $|\rho|$ drops 20%).

4.13 Robustness to Simulated Editing

Table 14 tests robustness to simulated human editing.

The depth signal degrades gracefully under simulated editing. LLM-simulated rephrasing ($\sim$1.7 sentences replaced per document) produces comparable degradation to 5% random word replacement (-4.5pp each). Even at 20% random replacement (F1 = 0.591), classification remains substantially above chance. Surface features are most vulnerable to word-level perturbation, while reference features are more robust because sentence-level perplexity is less sensitive to individual token replacements.

4.14 End-to-End Classifier Comparison

Table 15 compares the lightweight 7-feature ensemble against QLoRA fine-tuned end-to-end classifiers.

QLoRA classifiers achieve substantially higher within-distribution performance (best 3-class F1 =

Method	5-cl. F1	3-cl. F1	Δ
XGBoost	0.500	0.762	−0.262
CORAL (ordinal)	0.500	—	—
Frank-Hall (ordinal)	0.490	—	—

Table 12: Ordinal granularity comparison. 5-class F1 drops to ∼0.50 regardless of method. Adjacent-depth confusion rates: gen-1/gen-2 44%, gen-2/gen-3 53%, gen-3/gen-4 59%.

Generator	Size	Bin. F1	hpx ρ	ent ρ
Pythia-1B	1B	0.856	−0.63	−0.60
Qwen3-0.6B	0.6B	0.824	−0.58	−0.56
Pythia-2.8B	2.8B	0.804	−0.61	−0.60
Qwen3-1.7B	1.7B	0.805	−0.59	−0.55
OLMo-2-1B	1.5B	0.764	−0.60	−0.50
Qwen3-8B	8B	0.683	−0.56	−0.48

Table 13: Scale effect (6-gen LOGO held-out results). Qwen3-8B shows moderate degradation (0.683) but DDR structure is qualitatively preserved. 6-gen LOGO average: 0.789.

0.907), but are evaluated under within-generator CV, not LOGO—a fundamentally easier setting. End-to-end models are expected to degrade substantially under LOGO due to overfitting to generator-specific artifacts. The 7-feature ensemble's advantages are interpretability, efficiency (CPU-only, 1× cost vs 12× with 8.5GB GPU), and demonstrated cross-generator generalization under LOGO. Even with QLoRA, 5-class F1 remains 0.540–0.598, confirming the resolution ceiling is feature-intrinsic.

4.15 Text Length Confound Analysis

Text length is not a confound for depth classification (Table 16). Depth-length Spearman correlations are weak ($|\rho| < 0.14$). After length normalization, TTR and hapax depth drift are preserved and strengthened. The 4.6pp F1 drop under length-matched subsampling is attributable to reduced sample size (random subsample control at the same n yields ∼0.730), not confound removal.

4.16 LD-Score as Additional Feature

Adding LD-Score (a character-level distribution signature) as an 8th feature provides only +0.6pp marginal gain (Table 17), reflecting high redundancy with hapax ratio ($r = 0.75$). LD-Score's strong ordinal signal ($\rho = -0.79$) confirms that depth-induced distributional drift manifests at the character level, but its information is largely captured by existing surface features. We retain the 7-feature design for parsimony.

Edit Type	Bin. F1	Drop	95% CI
None (baseline)	0.695	—	[0.679, 0.711]
Random 5%	0.650	−4.5	[0.633, 0.667]
Random 10%	0.614	−8.1	[0.596, 0.632]
Random 20%	0.591	−10.4	[0.573, 0.609]
LLM rephrase	0.647	−4.5	[0.631, 0.662]

Table 14: Robustness to simulated editing. The signal degrades gracefully: −4.5pp at 5% random or LLM rephrase, −10.4pp at 20% random. Even at 20% replacement, F1 remains above random baseline.

Method	Eval	3-cl.	Bin.	Cost
7-feat XGBoost	LOGO	0.762	0.694	1×
QLoRA Qwen-Inst	CV	0.907	0.867	12×
QLoRA OLMo-Inst	CV	0.888	0.836	12×
QLoRA Qwen-Base	CV	0.904	0.860	12×
QLoRA OLMo-Base	CV	0.859	0.793	12×

Table 15: End-to-end (QLoRA) vs handcrafted features. QLoRA is evaluated under within-generator CV (not LOGO), making direct comparison unfair. QLoRA 5-class F1 ranges 0.540–0.598, confirming the resolution ceiling is feature-intrinsic, not method-limited.

5 Conclusion

We formulated generation depth classification as a task beyond binary AI-text detection and characterized differential depth response (DDR) across 5 base-model generators from 3 families. A lightweight 7-feature ensemble achieves leak-free binary F1 = 0.694 under LOGO evaluation, ranging 0.656–0.724 per generator, and 0.830 on cross-model chains. DDR is robust across tested decoding strategies (greedy, top-k) and three reference models from different families.

Our most robust finding is negative: depth-aware data curation is statistically equivalent to random subsampling for downstream CLM quality. Depth fingerprinting is diagnostic—useful for provenance characterization—not curative for data selection.

The signal degrades substantially under instruction tuning (OLMo-Instruct LOGO F1 = 0.338 vs 0.763 base), restricting current scope to base models. This degradation is itself a finding: instruction tuning compresses the distributional variation DDR exploits. Future work should investigate whether alternative feature families can recover depth signal in instruct models, extend evaluation to larger frontier models (>14B), and incorporate additional decoding strategies (nucleus sampling, beam search) to strengthen robustness claims.

Analysis	Result	Conclusion
ρ(length, depth)	$[-0.07, +0.13]$	Weak ($\|\rho\| < 0.14$)
Length-norm. drift	preserved, stronger	Depth $\perp$ length
Len-matched F1	0.724 vs 0.770	Sample-size effect
Rand. subsample	$\sim$0.730 (same n)	Confirms above

Table 16: Length confound analysis. Text length is not a confound: depth-length correlation is weak, and length normalization strengthens feature drift.

Feature Set	LOGO Bin. F1	Δ
7 features	0.694	—
7 features + LD-Score	0.700	$+0.6$
LD-Score alone	0.721	—

Table 17: LD-Score as additional feature. Marginal gain (+0.6pp) reflects high redundancy with hapax ratio ($r = 0.75$). LD-Score alone achieves 0.721 LOGO F1 but with wide cross-generator variance [0.588, 0.770].

Limitations

Our study has several limitations. First, all experiments use base models; instruction-tuned models show severely degraded signal, and extending depth fingerprinting to the instruct regime remains open. Second, our decoding robustness analysis covers greedy and top-k sampling; claims should be considered preliminary until validated on nucleus sampling and beam search. Third, the downstream utility experiment uses only perplexity on a single CLM task with 3 seeds—additional tasks and metrics are needed to confirm the "diagnostic not curative" conclusion broadly.